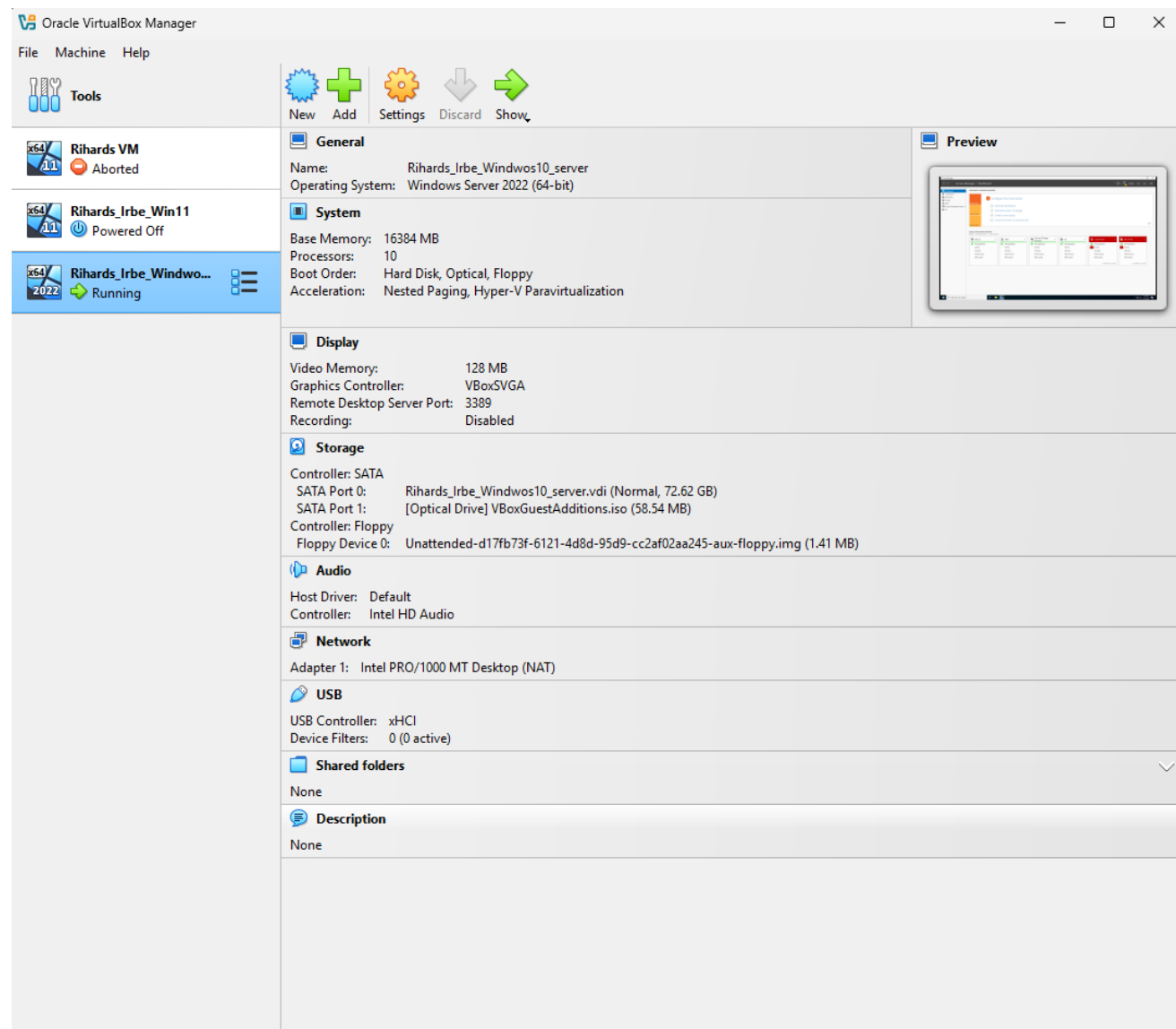


Operētājsistēmas

2. Praktiskais Darbs

Rihards Irbe 251RDB259(RDBG0 – 5.grupa)

Uz VirtualBox uzinstalēts Windows
2022 serveris un dators nosaukts:
"Rihards_Irbe_Windows10_server"





- Dashboard
- Local Server
- All Servers
- File and Storage Services ▸

WELCOME TO SERVER MANAGER

QUICK START

WHAT'S NEW

LEARN MORE

- 1 Configure this local server
- 2 Add roles and features
- 3 Add other servers to manage
- 4 Create a server group
- 5 Connect this server to cloud services

ROLES AND SERVER GROUPS

Roles: 1 | Server groups: 1 | Servers total: 1

File and Storage Services 1

Manageability

Events

Performance

BPA results

Local Server 1

Manageability

4 Events

5 Services

Performance

BPA results

11/23/2025 5:17 PM

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
Rihards-Irbe-Wi

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

DNS Server

Web Server Role (IIS)

Role Services

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
Rihards-Irbe-Wi	169.254.232.17...	Microsoft Windows Server 2022 Standard Evaluation

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous

Next >

Install

Cancel

Hide

FileMachineViewInputDevicesHelp

Server Manager

Server Manager ▸ Dashboard

Dashboard

Local Server

All Servers

AD DS

DNS

File and Storage Services ▸

IIS

WELCOME TO SERVER MANAGER

QUICK START

WHAT'S NEW

LEARN MORE

1

Configure this local server

2

Add roles and features

3

Add other servers to manage

4

Create a server group

5

Connect this server to cloud services

ROLES AND SERVER GROUPS

Roles: 4 | Server groups: 1 | Servers total: 1

AD DS1

Manageability

Events

Services

Performance

BPA results

DNS1

Manageability

Events

Services

Performance

BPA results

Add Roles and Features Wizard

Installation progress

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

DNS Server

Web Server Role (IIS)

Role Services

Confirmation

Results

View installation progress

Feature installation

Configuration required. Installation succeeded on Rihards-Irbe-Wi.

Active Directory Domain Services

Additional steps are required to make this machine a domain controller.

Promote this server to a domain controller

DNS Server

Group Policy Management

Remote Server Administration Tools

Role Administration Tools

AD DS and AD LDS Tools

Active Directory module for Windows PowerShell

AD DS Tools

Active Directory Administrative Center

You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.

Export configuration settings

< Previous

Next >

Close

Cancel

DESTINATION SERVER

Rihards-Irbe-Wi

Hide

All Servers1

Manageability

Events

Services

Performance

BPA results

11/23/2025 5:25 PM

- Pēc uzinstalēšanas, tika uzspiests uz Server Manager augšā karodziņš ar izsaukuma zīmi un uzinstalēts “Deployment Configuration”. Root domain name nosaukts kā “IRBER.local” un serveris kā “IRBER”, pēc uzdevuma nosacījumiem. Pēc tam tika restartēts dators un konfigurēts serveris. Tika veikti arī Windows atjauninājumi:

- Dashboard
- Local Server
- All Servers
- AD DS
- DNS
- File and Storage Services
- IIS

WELCOME TO SERVER MANAGER

1 Configure this local server

Hide

Active Directory Domain Services Configuration Wizard

Deployment Configuration

TARGET SERVER
Rihards-Irbe-Wi

Deployment Configuration

- Domain Controller Options
 - DNS Options
- Additional Options
- Paths
- Review Options
- Prerequisites Check
- Installation
- Results

Select the deployment operation

- ☐ Add a domain controller to an existing domain
- ☐ Add a new domain to an existing forest
- ☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous Next > Install Cancel

1

IIS 1

- Manageability
- Events
- Services
- Performance
- BPA results

1

Local Server 1

- Manageability
- Events 5
- Services 5
- Performance
- BPA results

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1

All Servers 1

- Manageability
- Events 5
- Services 5
- Performance
- BPA results

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Dashboard

- Local Server
- All Servers
- AD DS
- DNS
- File and Storage Services
- IIS

WELCOME TO SERVER MANAGER



Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
Rihards-Irbe-Wi

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

Hide

1

IIS 1

- Manageability
- Events
- Services
- Performance
- BPA results

Local Server 1

- Manageability
- Events 5
- Services 5
- Performance
- BPA results

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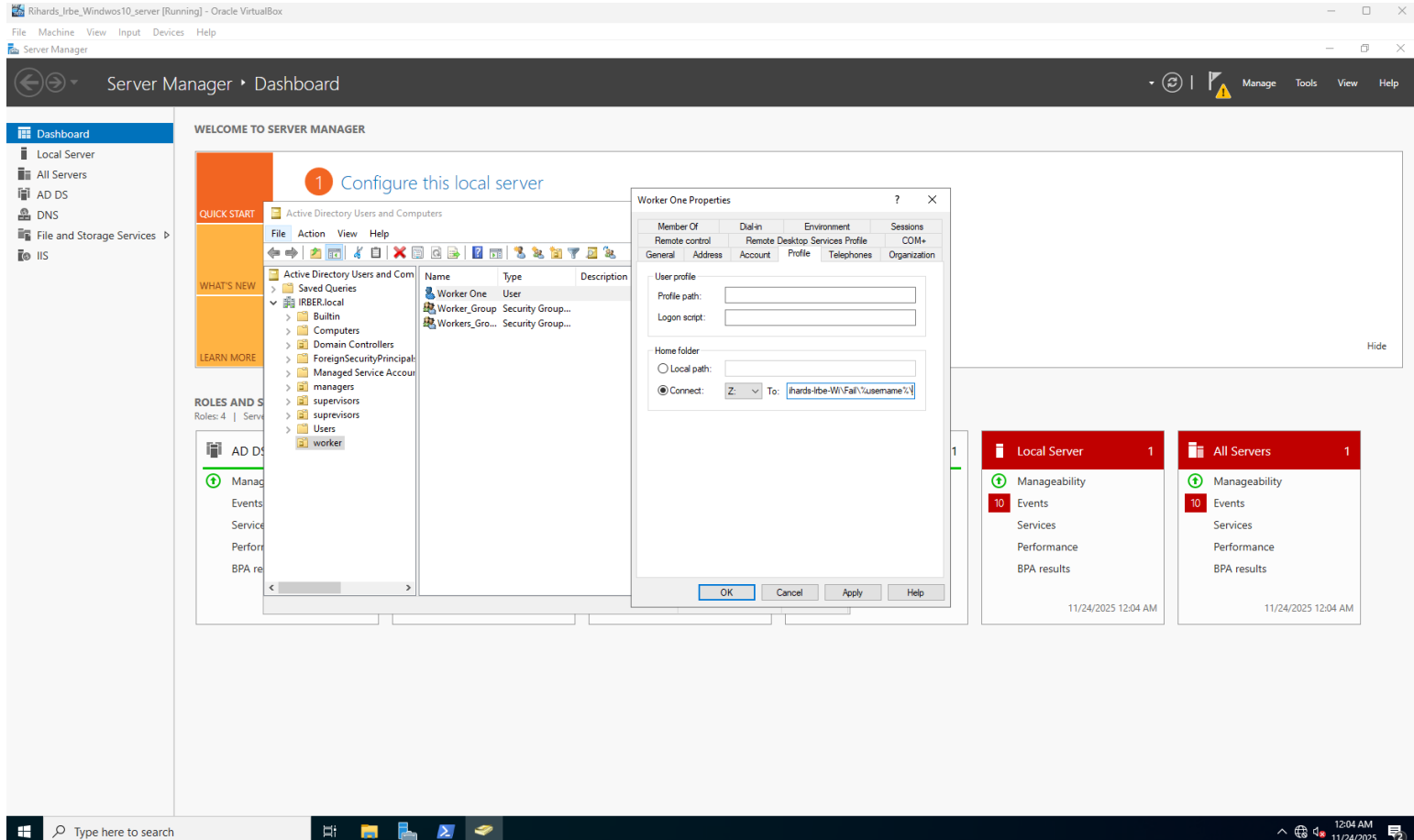
All Servers 1

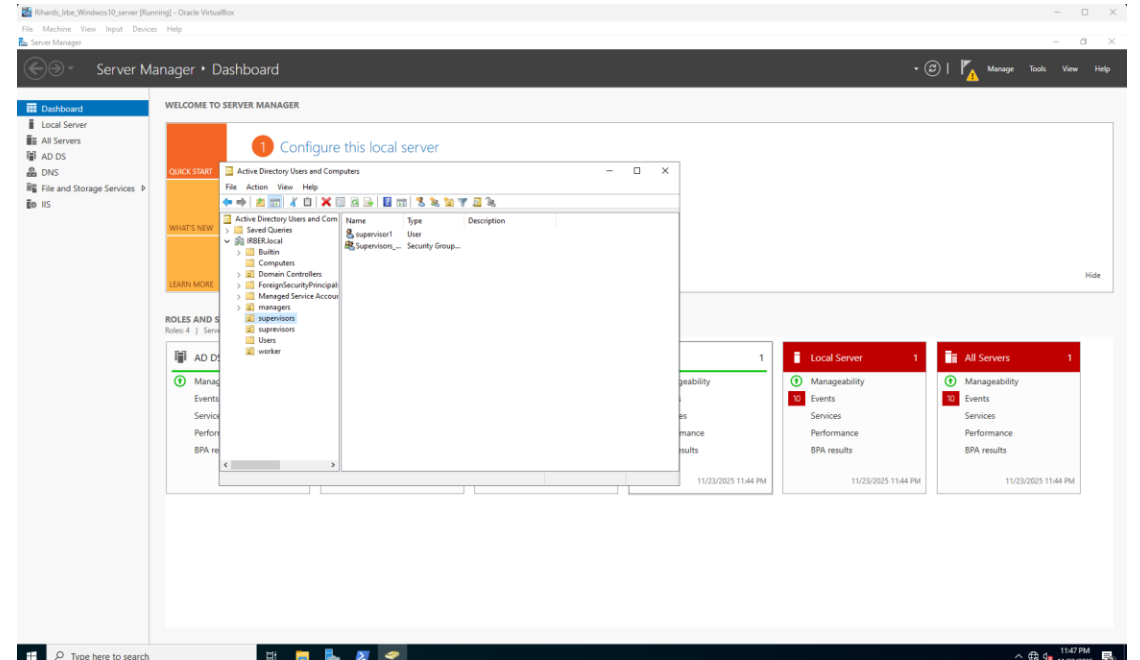
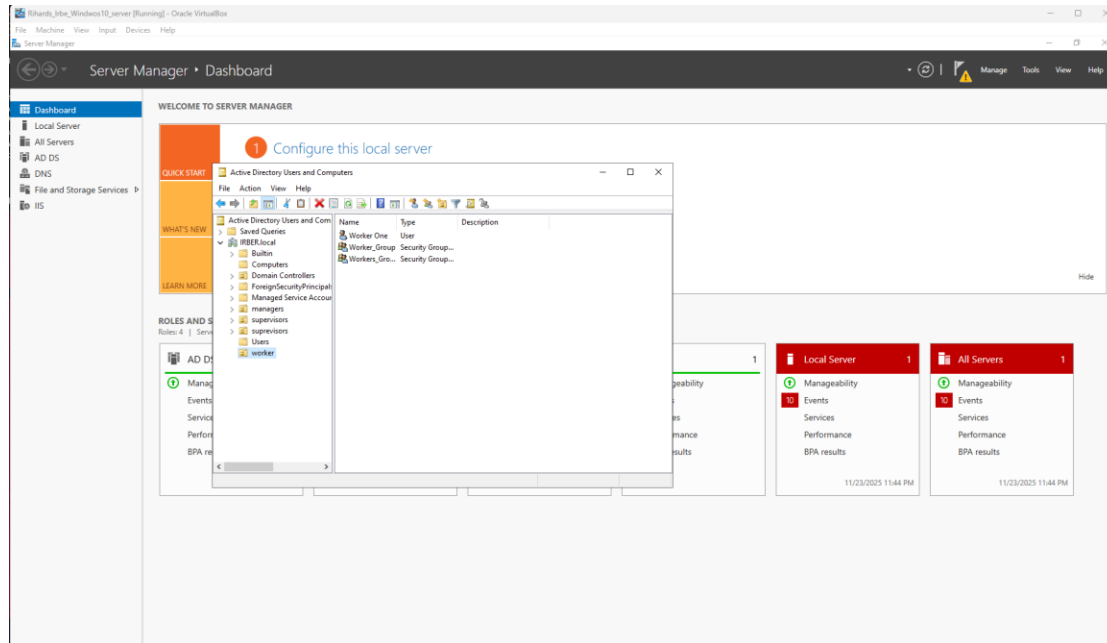
- Manageability
- Events 5
- Services 5
- Performance
- BPA results

11/23/2025 5:31 PM

2.

- **Aktīvā katalogā izveidotas organizatoriskās vienības, globālās drošības grupas un lietotāji ar mājas mapēm.**
Katram lietotājam mājas mape atrodas diskā un tiek automātiski pievienota kā Home Folder.
Organizatoriskās vienības sadala lietotājus pēc lomas (Managers, Workers, Supervisors), grupas nodrošina piekļuves tiesības, un mapes tiek aizsargātas, lai tikai attiecīgais lietotājs un administratori varētu modificēt saturu.







Dashboard

- Local Server
- All Servers
- AD DS
- DNS
- File and Storage Services ▸
- IIS

WELCOME TO SERVER MANAGER

QUICK START

WHAT'S NEW

LEARN MORE

ROLES AND SERVICES

Roles: 4 | Services: 1

AD DS



Manageability

Events

Services

Performance

BPA results

1 Configure this local server

Active Directory Users and Computers

File Action View Help

Active Directory Users and Computers

Saved Queries

IRBER.local

Builtin

Computers

Domain Controllers

ForeignSecurityPrincipal...

Managed Service Accounts

managers

supervisors

suprevisors

Users

worker

Name	Type	Description
Managers_G...	Security Group...	
Test One	User	
Test Two	User	

Hide

1

Local Server 1

Manageability

10 Events

Services

Performance

BPA results

11/23/2025 11:44 PM

All Servers 1

Manageability

10 Events

Services

Performance

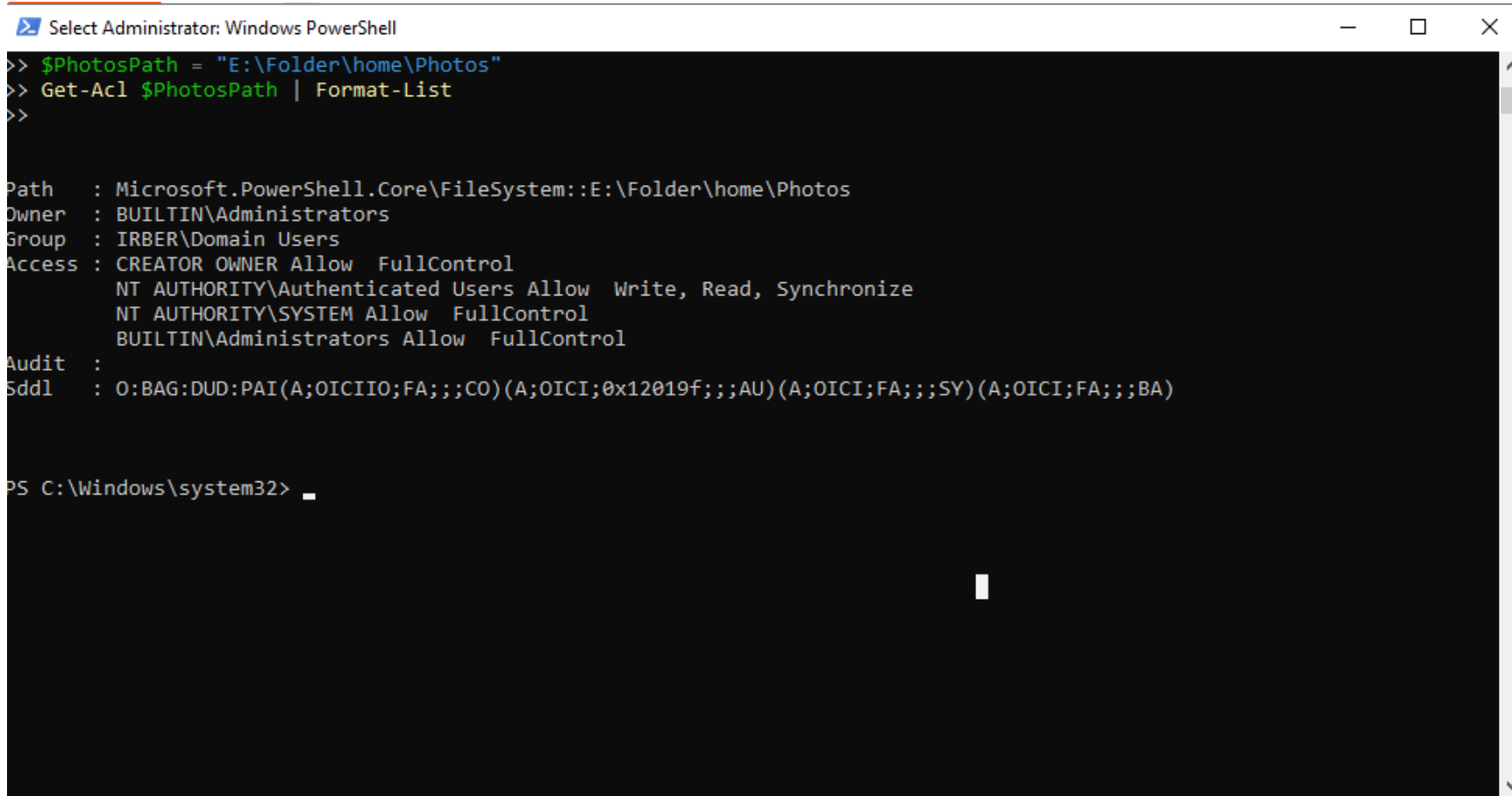
BPA results

11/23/2025 11:44 PM

3) Koplietojuma resursi

- Diskā **E**: izveidotas galvenās koplietojuma mapes:
- **Users** – katram lietotājam sava mājas mape, piekļuve tikai konkrētajam lietotājam, administratoriem un sistēmai.
- **RegulatoryDocs** – organizācijas reglamentējošie dokumenti, parastie lietotāji var tikai skatīt un kopēt failus, administratori pilna piekļuve.
- **Photos** – visi lietotāji var lasīt un kopēt failus, bet rediģēt un dzēst drīkst tikai faila autors, izmantojot **Creator Owner** tiesības.
- **ManagementOnly** – piekļuve tikai grupai **Managers_Group**, failus mapē iekopē sekretāre, vadība var pilnībā rediģēt saturu.

Permisiju sarkasts ar konsoli



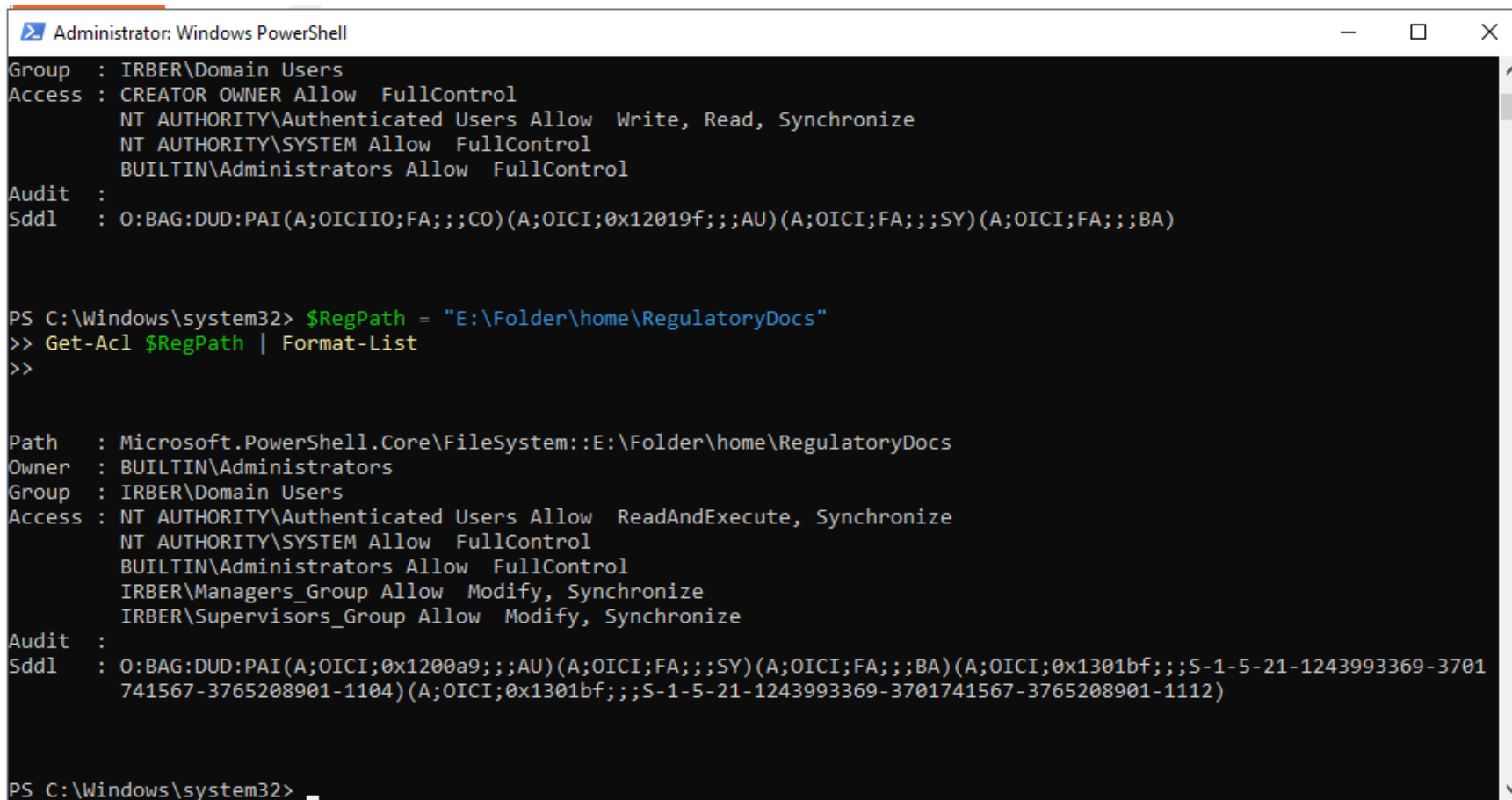
```
Select Administrator: Windows PowerShell

>> $PhotosPath = "E:\Folder\home\Photos"
>> Get-Acl $PhotosPath | Format-List
>>

Path      : Microsoft.PowerShell.Core\FileSystem::E:\Folder\home\Photos
Owner     : BUILTIN\Administrators
Group     : IRBER\Domain Users
Access    : CREATOR OWNER Allow FullControl
           NT AUTHORITY\Authenticated Users Allow Write, Read, Synchronize
           NT AUTHORITY\SYSTEM Allow FullControl
           BUILTIN\Administrators Allow FullControl
Audit     :
Sddl      : O:BAG:DUD:PAI(A;OICIIO;FA;;;CO)(A;OICI;0x12019f;;;AU)(A;OICI;FA;;;SY)(A;OICI;FA;;;BA)

PS C:\Windows\system32>
```

Permisiju sarkasts ar konsoli



```
Administrator: Windows PowerShell

Group : IRBER\Domain Users
Access : CREATOR OWNER Allow FullControl
        NT AUTHORITY\Authenticated Users Allow Write, Read, Synchronize
        NT AUTHORITY\SYSTEM Allow FullControl
        BUILTIN\Administrators Allow FullControl
Audit :
Sddl : O:BAG:DUD:PAI(A;OICIIO;FA;;;CO)(A;OICI;0x12019f;;;AU)(A;OICI;FA;;;SY)(A;OICI;FA;;;BA)

PS C:\Windows\system32> $RegPath = "E:\Folder\home\RegulatoryDocs"
>> Get-Acl $RegPath | Format-List
>>

Path      : Microsoft.PowerShell.Core\FileSystem::E:\Folder\home\RegulatoryDocs
Owner     : BUILTIN\Administrators
Group     : IRBER\Domain Users
Access    : NT AUTHORITY\Authenticated Users Allow ReadAndExecute, Synchronize
           NT AUTHORITY\SYSTEM Allow FullControl
           BUILTIN\Administrators Allow FullControl
           IRBER\Managers_Group Allow Modify, Synchronize
           IRBER\Supervisors_Group Allow Modify, Synchronize
Audit     :
Sddl      : O:BAG:DUD:PAI(A;OICI;0x1200a9;;;AU)(A;OICI;FA;;;SY)(A;OICI;FA;;;BA)(A;OICI;0x1301bf;;;S-1-5-21-1243993369-3701741567-3765208901-1104)(A;OICI;0x1301bf;;;S-1-5-21-1243993369-3701741567-3765208901-1112)

PS C:\Windows\system32>
```

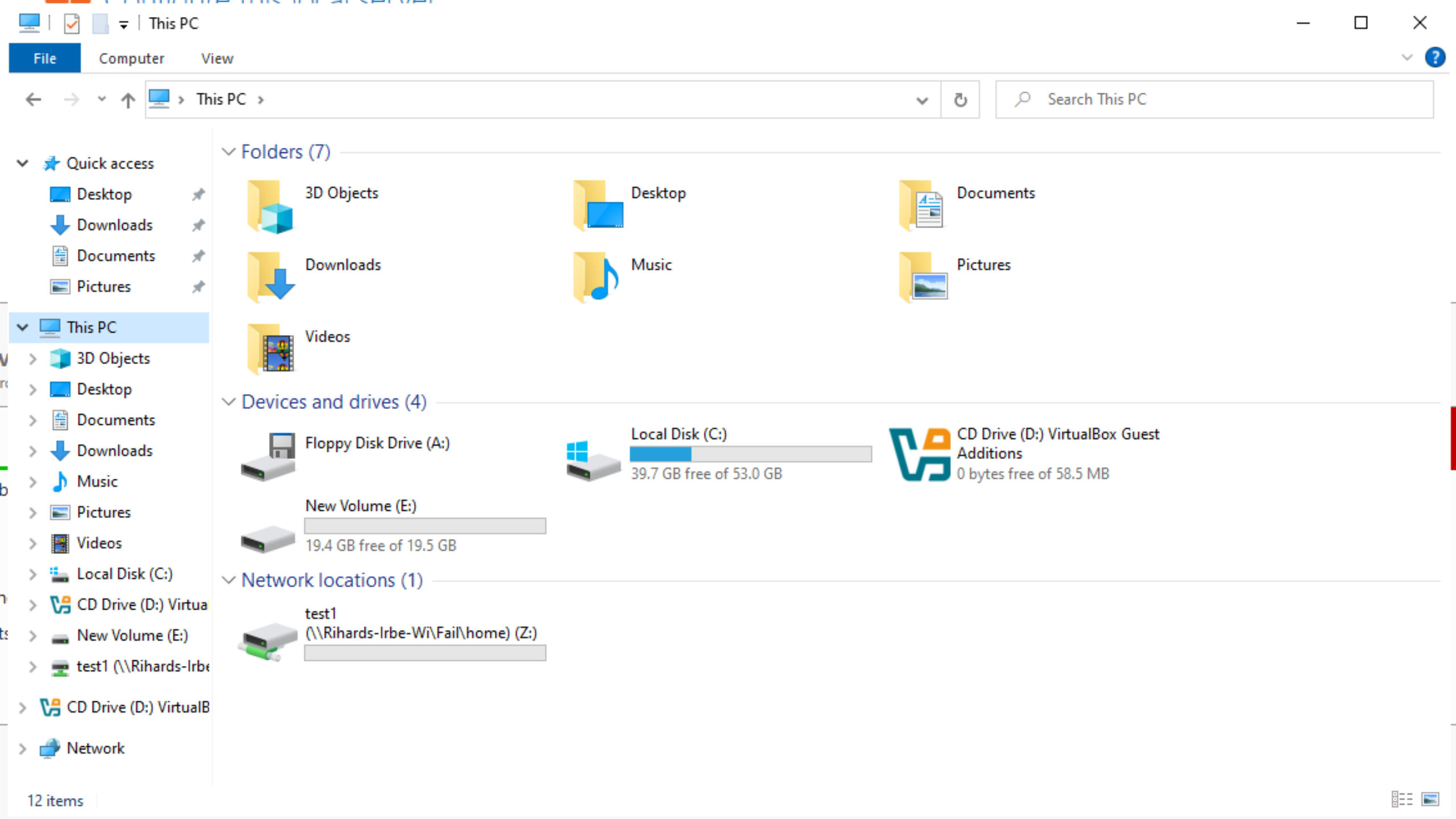
Administrator: Windows PowerShell

```
Owner : BUILTIN\Administrators
Group : IRBER\Domain Users
Access : NT AUTHORITY\Authenticated Users Allow ReadAndExecute, Synchronize
        NT AUTHORITY\SYSTEM Allow FullControl
        BUILTIN\Administrators Allow FullControl
        IRBER\Managers_Group Allow Modify, Synchronize
        IRBER\Supervisors_Group Allow Modify, Synchronize
Audit :
Sddl : O:BAG:DUD:PAI(A;OICI;0x1200a9;;;AU)(A;OICI;FA;;;SY)(A;OICI;FA;;;BA)(A;OICI;0x1301bf;;;S-1-5-21-1243993369-3701741567-3765208901-1104)(A;OICI;0x1301bf;;;S-1-5-21-1243993369-3701741567-3765208901-1112)

PS C:\Windows\system32> $MgmtPath = "E:\Folder\home\ManagementOnly"
>> Get-Acl $MgmtPath | Format-List
>>

Path : Microsoft.PowerShell.Core\FileSystem::E:\Folder\home\ManagementOnly
Owner : BUILTIN\Administrators
Group : IRBER\Domain Users
Access : NT AUTHORITY\SYSTEM Allow FullControl
        BUILTIN\Administrators Allow FullControl
        IRBER\Managers_Group Allow FullControl
Audit :
Sddl : O:BAG:DUD:PAI(A;OICI;FA;;;SY)(A;OICI;FA;;;BA)(A;OICI;FA;;;S-1-5-21-1243993369-3701741567-3765208901-1104)

PS C:\Windows\system32>
```



Domēna un tā daļas lietotāju politikas

Tālāk izmainīju arī grupu “lockout” politikas.

Local Server

All Servers

AD DS

DNS

File and Storage

IIS

Group Policy Management

File Action View Window Help

Group Policy Management

Forest: IRBER.local

Domains

IRBER.local

Default Domain

Domain Contro

managers

supervisors

suprevisors

worker

Group Policy Ol

WMI Filters

Starter GPOs

Sites

Group Policy Modeling

Group Policy Results

Group Policy Management

File Action View Window Help

Default Domain Policy

Scope Details Settings Delegation

Links

Display links in this location: IRBER.local

The following sites, domains, and OUs are linked to this GPO:

Location

Enforced

IRBER.local

No

Security Filtering

The settings in this GPO can only apply to the following groups, users,

Name

Authenticated Users

Add... Remove Properties

WMI Filtering

This GPO is linked to the following WMI filter:

Group Policy Management Editor

File Action View Help

Default Domain Policy [RIHARDS-IRBE-WI.IRBER.LOCAL] Policy

Computer Configuration

Policies

Software Settings

Windows Settings

Name Resolut

Scripts (Startu

Security Settin

Account P

Password

Accou

Kerber

Local Polic

Event Log

Restricted

System Se

Registry

File System

Wired Net

Windows I

Network L

Wireless N

Public Key

Software F

Policy Setting

Account lockout duration

15 minutes

Account lockout threshold

10 invalid logon attempts

Reset account lockout counter after

15 minutes

BPA results

BPA results

1

All Servers

1

Manageability

5 Events

1 Services

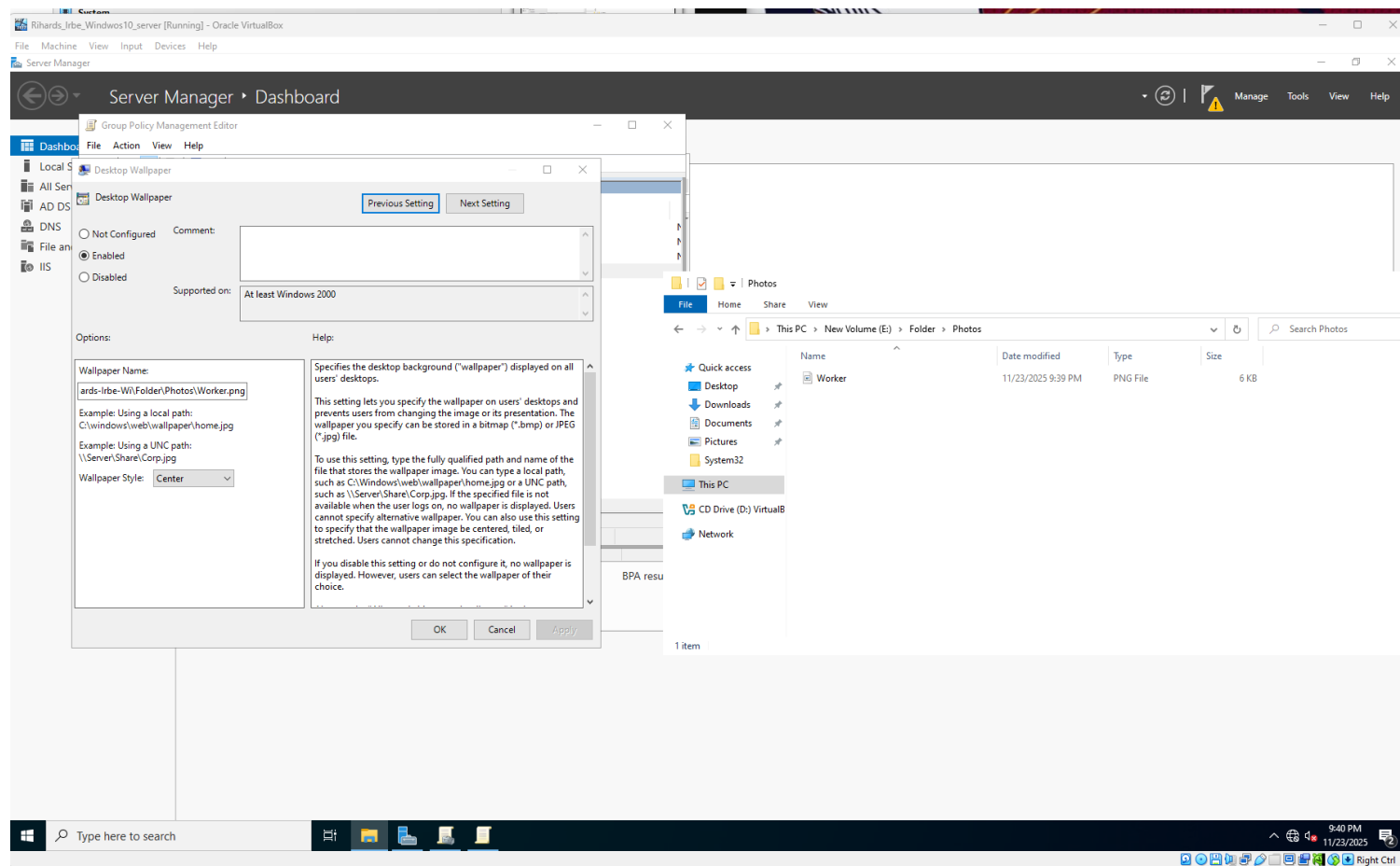
Performance

BPA results

11 PM

11/23/2025 9:31 PM

Samainīju darbinieku darbavirsmas tapeti



5) Tīmekļa serveris IIS ar vienkāršu 1 faila html lapu

- uzdevumā instalēju un konfigurēju **IIS (Internet Information Services)**, izmantojot PowerShell. Izveidoju vienkāršu **HTML testlapu** mapē `C:\inetpub\wwwroot`, kurā norādīts serveris, domēns un students. Noņēmu noklusējuma **Users grupas** tiesības un piešķīru pareizas piekļuves **IUSR** un **IIS_IUSRS** kontiem, nodrošinot drošu lapas darbību. Pārbaudīju IIS servisa statusu, lapas pieejamību, mapes atļaujas un HTML faila izveidi, izmantojot PowerShell skriptus.

```

Administrator: Windows PowerShell

>> <html>
>> <head>
>> <title>Rihards Webpage</title>
>> </head>
>> <body>
>> <h1>HTML fails strada</h1>
>> <p>Domain: IRBER.local</p>
>> <p>Student: Rihards Irbe (251RDB259)</p>
>> </body>
>> </html>
>>
>> "@ | Out-File -FilePath "C:\inetpub\wwwroot\index.html" -Encoding UTF8 -Force
>>
>> # Remove your write permissions after (for security)
>> icacls "C:\inetpub\wwwroot\" /remove "IRBER\rihards_irbe"
processed file: C:\inetpub\wwwroot\
Successfully processed 1 files; Failed processing 0 files
processed file: C:\inetpub\wwwroot\
Successfully processed 1 files; Failed processing 0 files
PS C:\inetpub\wwwroot> # Test the updated page
>> Start-Process "http://localhost"
>>
>> # Or check via PowerShell
>> Invoke-WebRequest -Uri "http://localhost" -UseBasicParsing | Select-Object StatusCode, Content

StatusCode Content
-----
200 I>>
PS C:\inetpub\
    
```

Rihards Webpage

localhost

HTML fails strada

Domain: IRBER.local

Student: Rihards Irbe (251RDB259)

Manage Tools View Help

Hide

All Servers 1

Manageability

Events

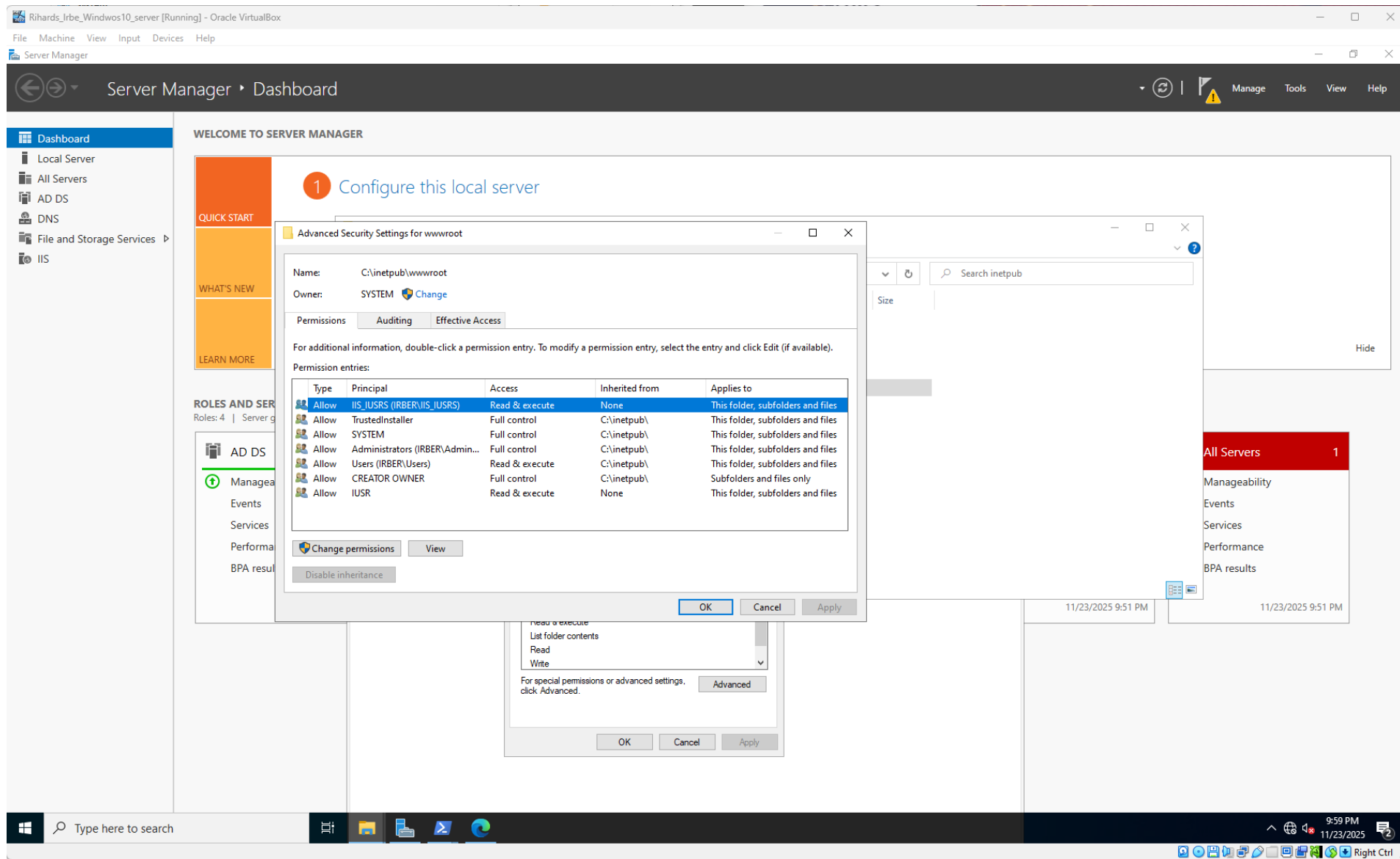
Services

Performance

BPA results

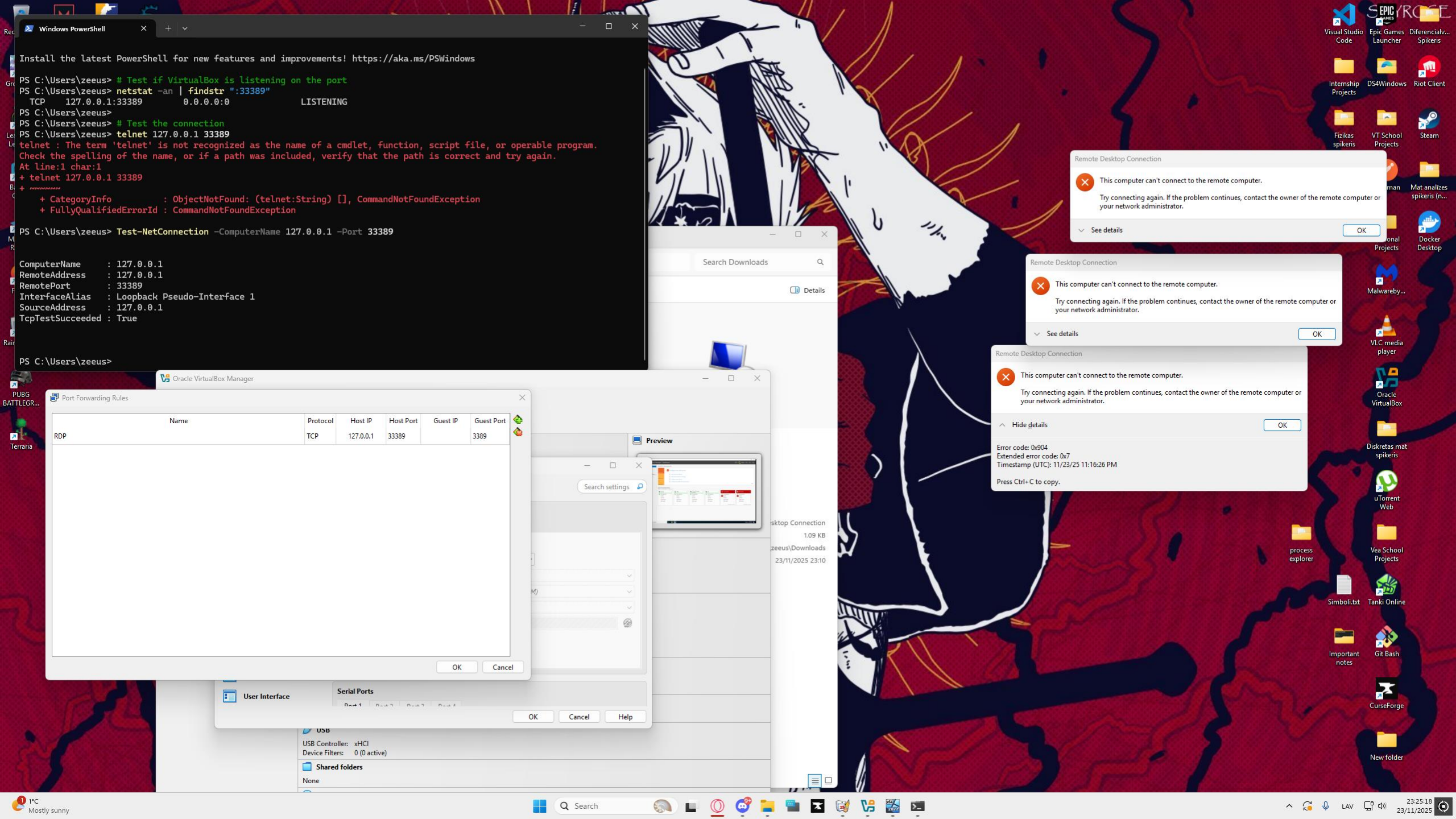
11/23/2025 9:51 PM

Ln 1, Col 1 100% Unix (LF) UTF-8 with BOM



6) Papilduzdevumi

- Centos izpildīt Remote Desktop, taču nesanāca. Pajautāju mākslīgajam intelektam un nonāca pie secinājuma kad man ir "corrupted" servera instalācija. Nezinu vai tā ir pateisība, bet patiešām nezināmu iemeslu dēļ nevarēju veiksmīgi pieslēgties ar remote desktop.



```
Windows PowerShell
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\zeus> # Test if VirtualBox is listening on the port
PS C:\Users\zeus> netstat -an | findstr ":33389"
TCP    127.0.0.1:33389    0.0.0.0:0        LISTENING
PS C:\Users\zeus>
PS C:\Users\zeus> # Test the connection
PS C:\Users\zeus> telnet 127.0.0.1 33389
telnet : The term 'telnet' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ telnet 127.0.0.1 33389
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (telnet:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\zeus> Test-NetConnection -ComputerName 127.0.0.1 -Port 33389

ComputerName      : 127.0.0.1
RemoteAddress     : 127.0.0.1
RemotePort        : 33389
InterfaceAlias    : Loopback Pseudo-Interface 1
SourceAddress     : 127.0.0.1
TcpTestSucceeded  : True

PS C:\Users\zeus>
```

Port Forwarding Rules

Name	Protocol	Host IP	Host Port	Guest IP	Guest Port
RDP	TCP	127.0.0.1	33389		3389

User Interface

Serial Ports

Port 1 Port 2 Port 3 Port 4

OK Cancel Help

Remote Desktop Connection

This computer can't connect to the remote computer.

Try connecting again. If the problem continues, contact the owner of the remote computer or your network administrator.

See details OK

Remote Desktop Connection

This computer can't connect to the remote computer.

Try connecting again. If the problem continues, contact the owner of the remote computer or your network administrator.

See details OK

Remote Desktop Connection

This computer can't connect to the remote computer.

Try connecting again. If the problem continues, contact the owner of the remote computer or your network administrator.

Hide details OK

Error code: 0x904
Extended error code: 0x7
Timestamp (UTC): 11/23/25 11:16:26 PM
Press Ctrl+C to copy.

Secinājumi

- Veicot uzdevumus, izdevās veiksmīgi konfigurēt Windows Server 2022 ar Active Directory, izveidot lietotāju un grupu struktūru, piešķirt pareizas mapju un koplietojumu atļaujas, kā arī uzstādīt IIS ar drošu HTML lapu. Visi uzstādījumi tika pārbaudīti un darbojas atbilstoši prasībām, nodrošinot drošu un organizētu servera vidi ar pārskatāmām piekļuves tiesībām. Vienīgās grūtības bija Remote Desktop, kuru pēc 3-4 stundām nevarēju piedabūt pie dzīvības